

Foundations of Data Science

Lecture 3: Visualization

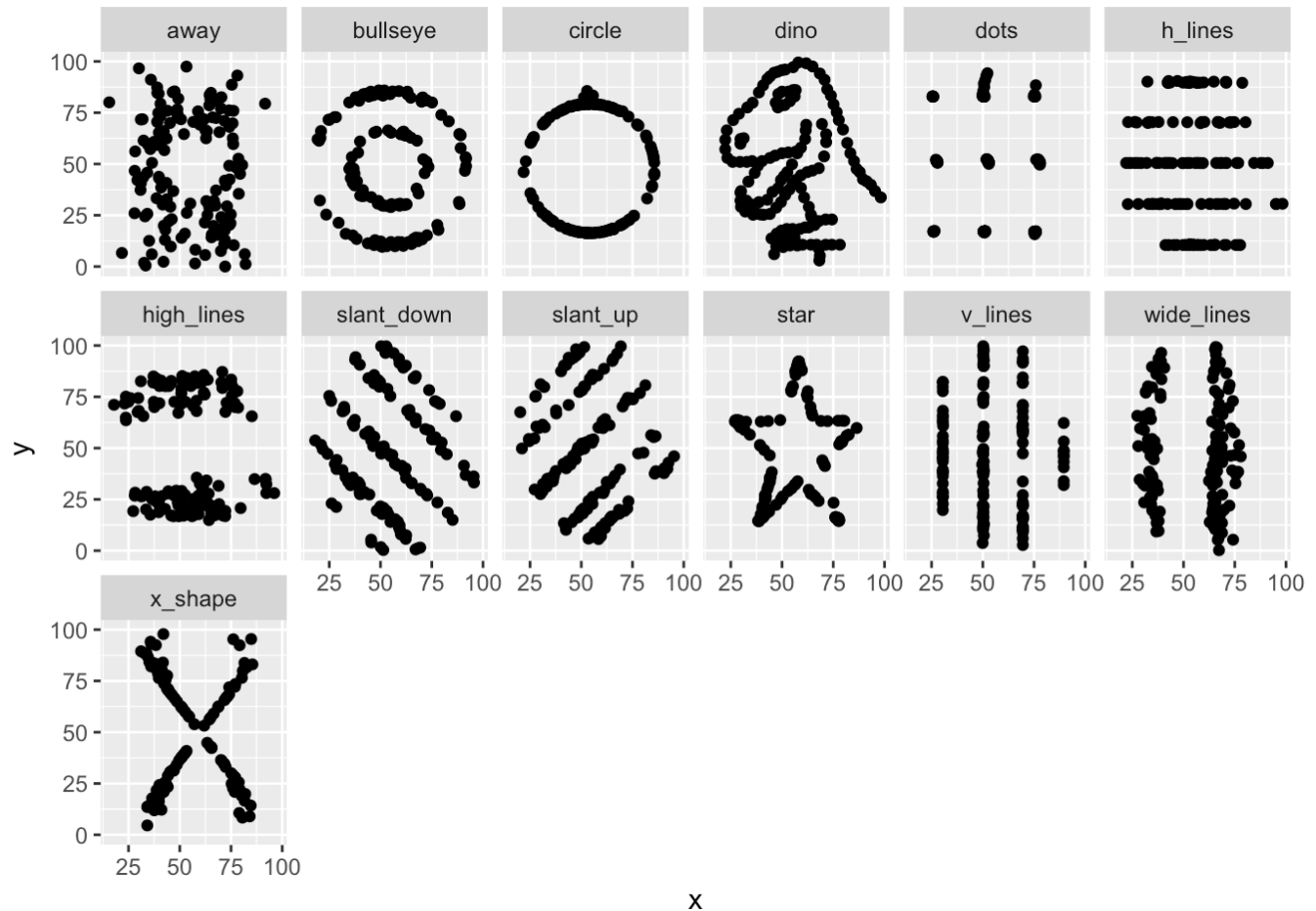
Prof. Gilles Louppe

g.louppe@uliege.be

dataset	mean_x	mean_y	sd_x	sd_y	cor
away	54.26610	47.83472	16.76982	26.93974	-0.06412835
bullseye	54.26873	47.83082	16.76924	26.93573	-0.06858639
circle	54.26732	47.83772	16.76001	26.93004	-0.06834336
dino	54.26327	47.83225	16.76514	26.93540	-0.06447185
dots	54.26030	47.83983	16.76774	26.93019	-0.06034144
h_lines	54.26144	47.83025	16.76590	26.93988	-0.06171484
high_lines	54.26881	47.83545	16.76670	26.94000	-0.06850422
slant_down	54.26785	47.83590	16.76676	26.93610	-0.06897974
slant_up	54.26588	47.83150	16.76885	26.93861	-0.06860921
star	54.26734	47.83955	16.76896	26.93027	-0.06296110
v_lines	54.26993	47.83699	16.76996	26.93768	-0.06944557
wide_lines	54.26692	47.83160	16.77000	26.93790	-0.06657523
x_shape	54.26015	47.83972	16.76996	26.93000	-0.06558334

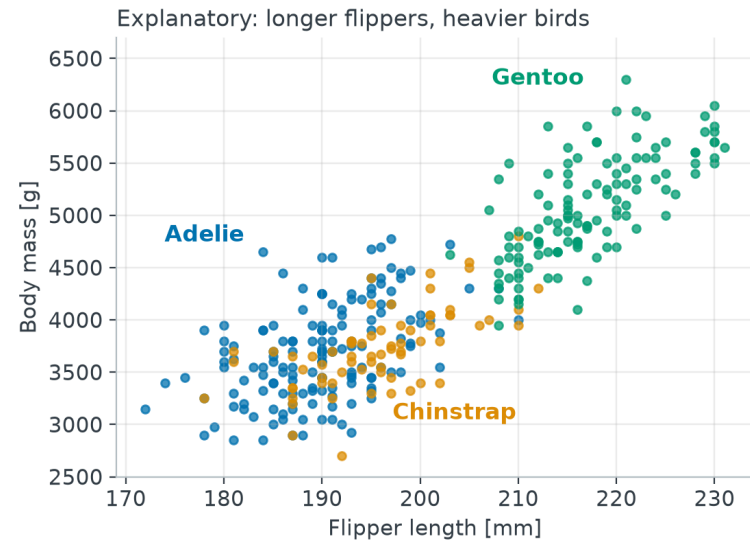
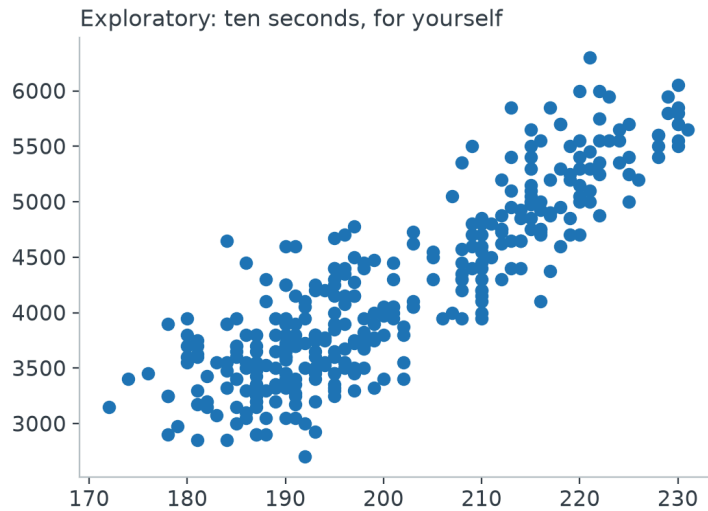
Do not summarize the data without visualizing it.

The Datasaurus dozen: 13 datasets with identical summary statistics.



Summary statistics can be misleading. **Always visualize the rawest data!**

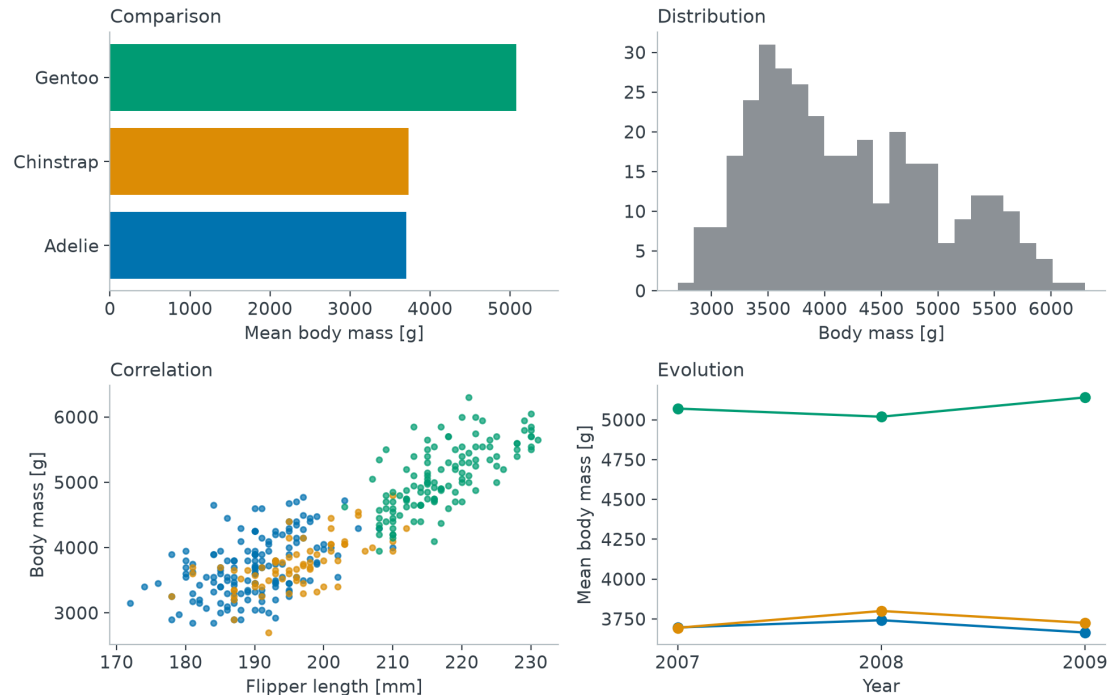
Two kinds of plots



- **Exploratory:** for yourself, quickly and in numbers, to see (Lecture 2).
- **Explanatory:** for others, few and polished, to carry one message.

Choosing the right plot

Start from the message



Choose the graph from the question it answers, then the design from that question and the number of continuous variables.

What that gives

- **Comparison:** horizontal bars, which must start at zero, or dots along a scale, which need not.
- **Distribution:** every point along a scale; a histogram or a box plot when there are too many.
- **Correlation:** a scatter plot, or an array of them beyond two variables.
- **Evolution:** lines against the independent variable; different units go in different panels sharing a scale.

This is one opinionated but consistent take, not the only one.

Choosing the right graph

Principiae



Watch on

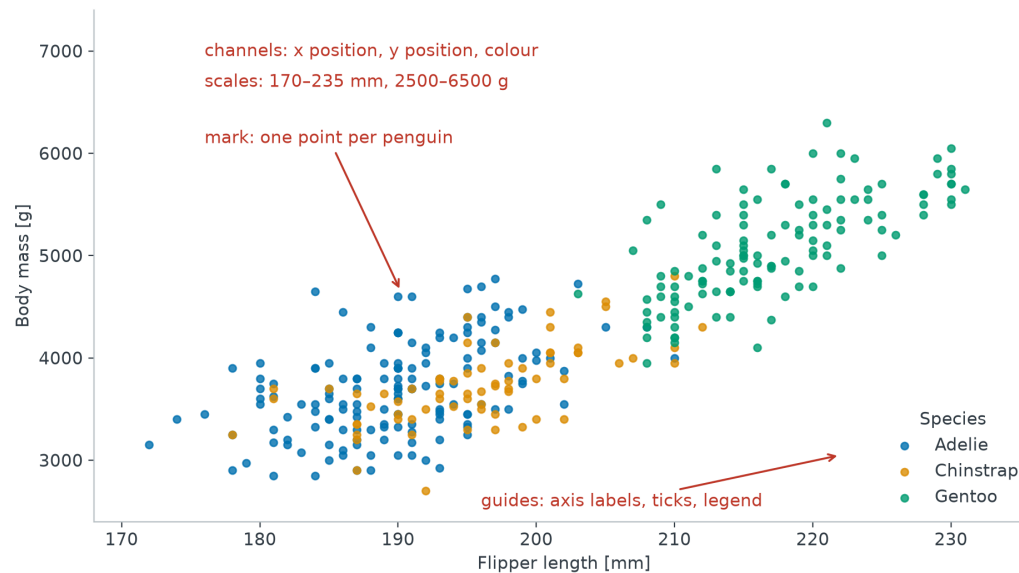
Jean-Luc Doumont, "Choosing the right graph", 2017.

Encoding data with visual cues

14, 37, 75

In pairs, try to come up with as many representations/encodings of this "data" as possible.

Anatomy of a plot



data → marks → channels → scales → coordinates → guides

Visual cues

Visual cues are elements of a visualization that encode data. They are expressed as marks and channels:

- A **mark** is a geometric primitive such as points, lines, or areas.
- A **channel** is an attribute of a mark that can be used to encode data, such as position, size, shape, or color.

➔ Points



➔ Lines



➔ Areas



Marks are geometric primitives.

→ Position

→ Horizontal



→ Vertical



→ Both



→ Color



→ Shape



→ Tilt

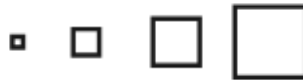


→ Size

→ Length



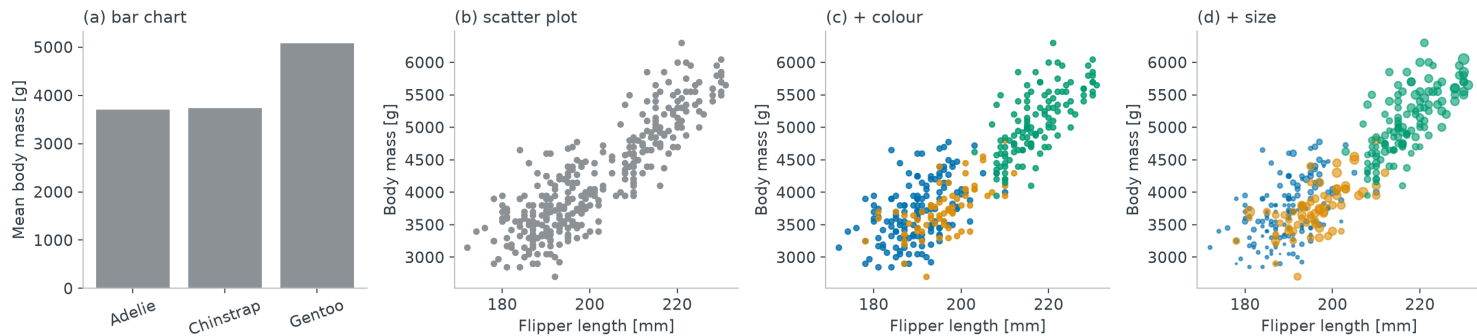
→ Area



→ Volume

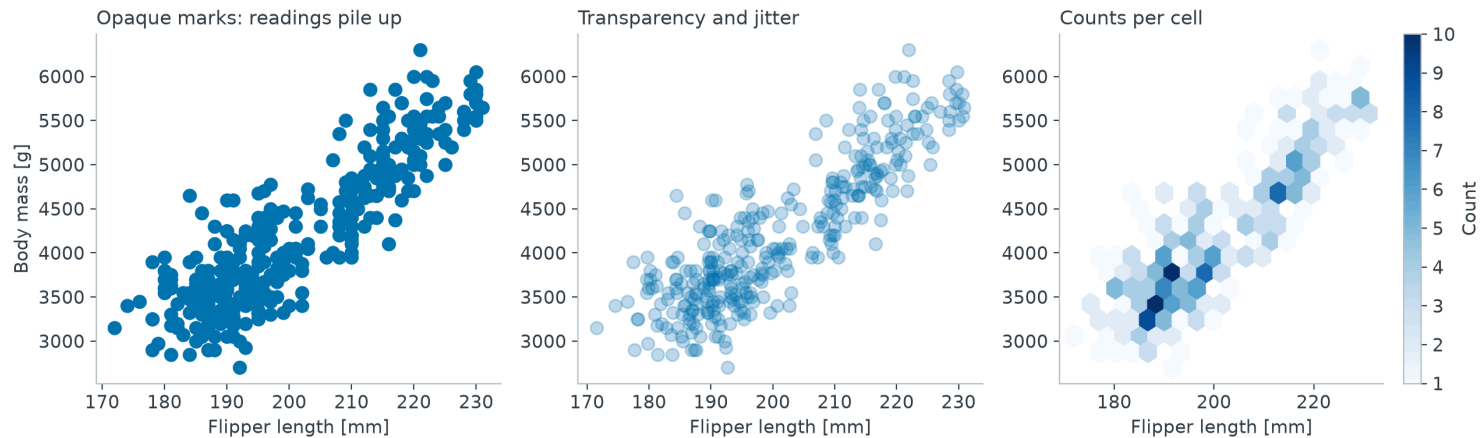


Visual channels control the appearance of marks.



(a) Bar charts encode data using line marks, controlled by a vertical position channel (height) and a horizontal position channel (category). (b) Scatterplots encode data using point marks, controlled by two position channels (x and y). (c) A third variable can be encoded using a color channel. (d) A fourth variable can be encoded using a size channel (area of the point).

Overplotting



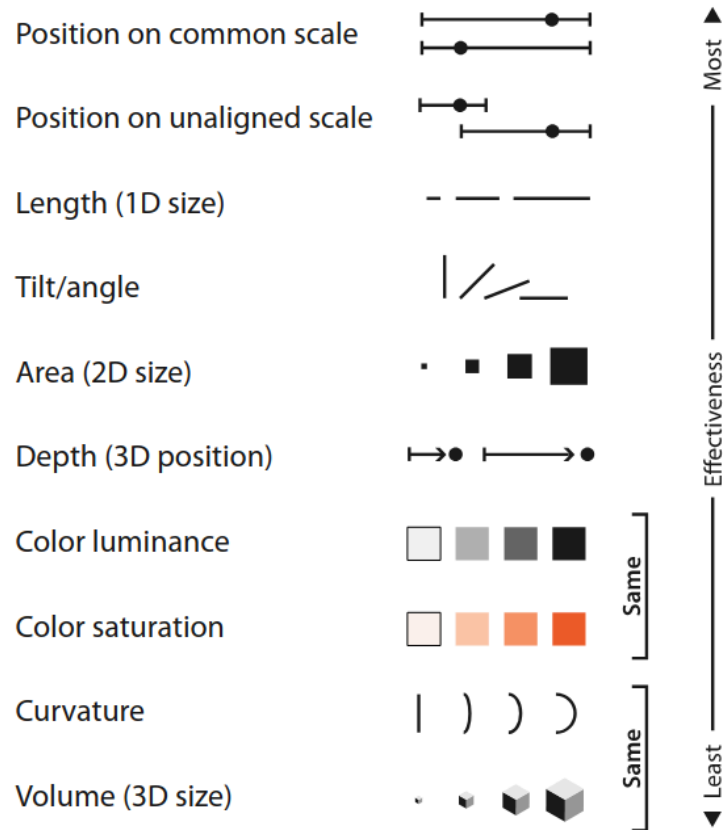
Flipper lengths are whole millimetres, body masses multiples of 25 g, so identical readings stack. Transparency, jitter or binning give the density back.

Two principles

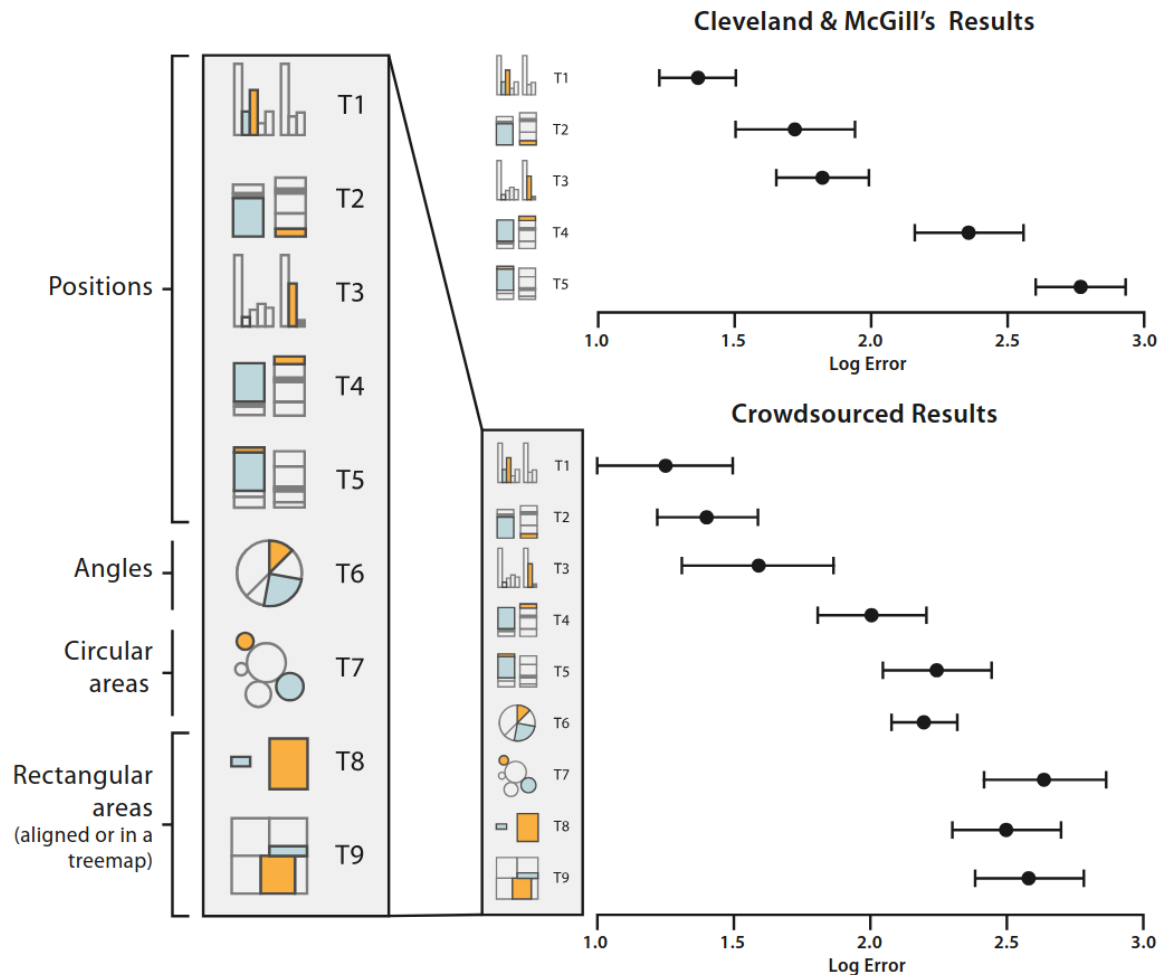
- **Expressiveness:** show all the data facts, and only the data facts.
- **Effectiveness:** encode them with the channels a reader decodes most accurately.

Perceptual hierarchy

Effectiveness is measurable: not all channels are read equally well.



Cleveland and McGill (1984) conducted experiments to evaluate the accuracy of visual channels for encoding quantitative data.



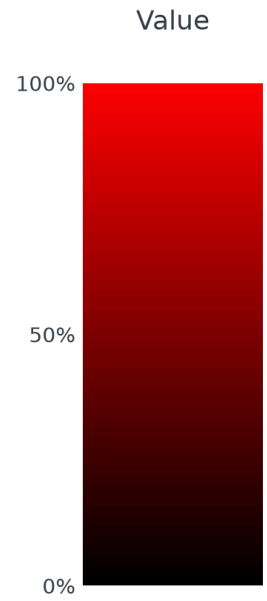
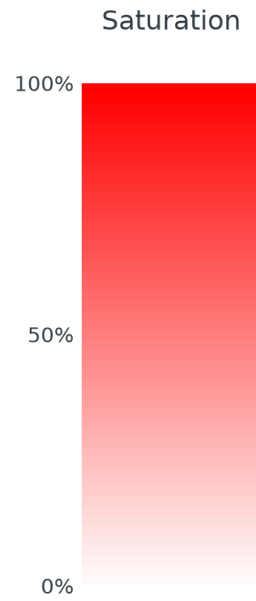


Colors

Color is a powerful channel for encoding categorical and quantitative data.

The primary representation system is the Hue, Saturation, Value (HSV) model:

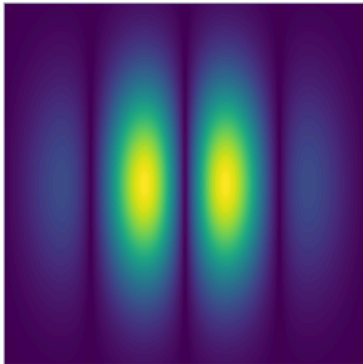
- **Hue**: the type of color (e.g., red, green, blue), numerically represented as an angle on the color wheel (0-360 degrees).
- **Saturation**: the intensity or purity of the color (from gray to full color), represented as a percentage (0-100%).
- **Value**: the brightness of the color (from black to full brightness), represented as a percentage (0-100%).



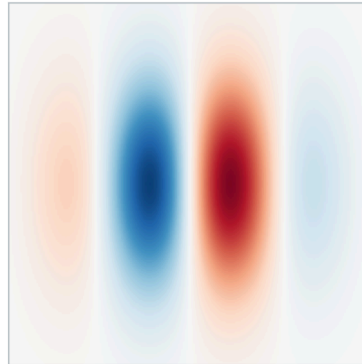
A colormap maps data values to colors. Three kinds, for three kinds of data:

- **Sequential**: ordered data, light to dark within a single hue.
- **Diverging**: ordered data with a meaningful midpoint, two hues.
- **Categorical**: unordered groups, distinct colors.

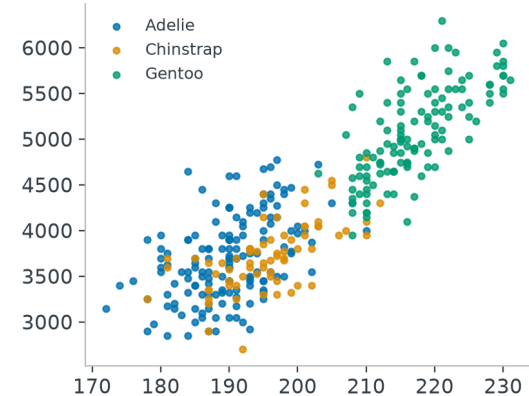
Sequential: ordered data

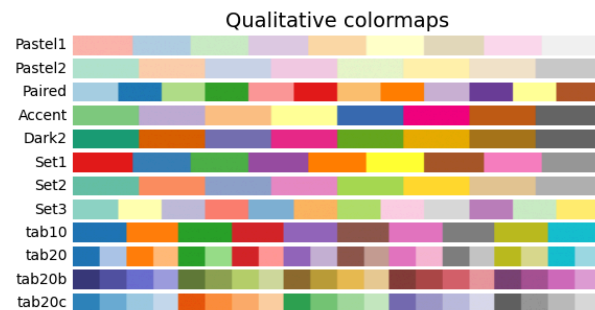
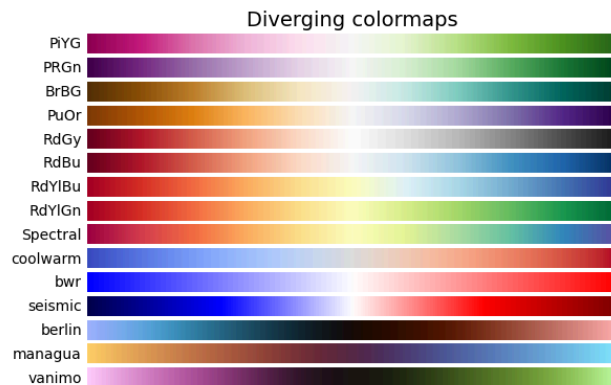
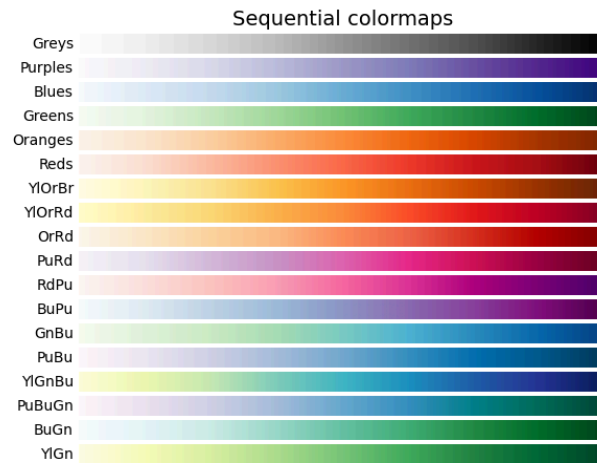


Diverging: a meaningful zero



Categorical: unordered groups



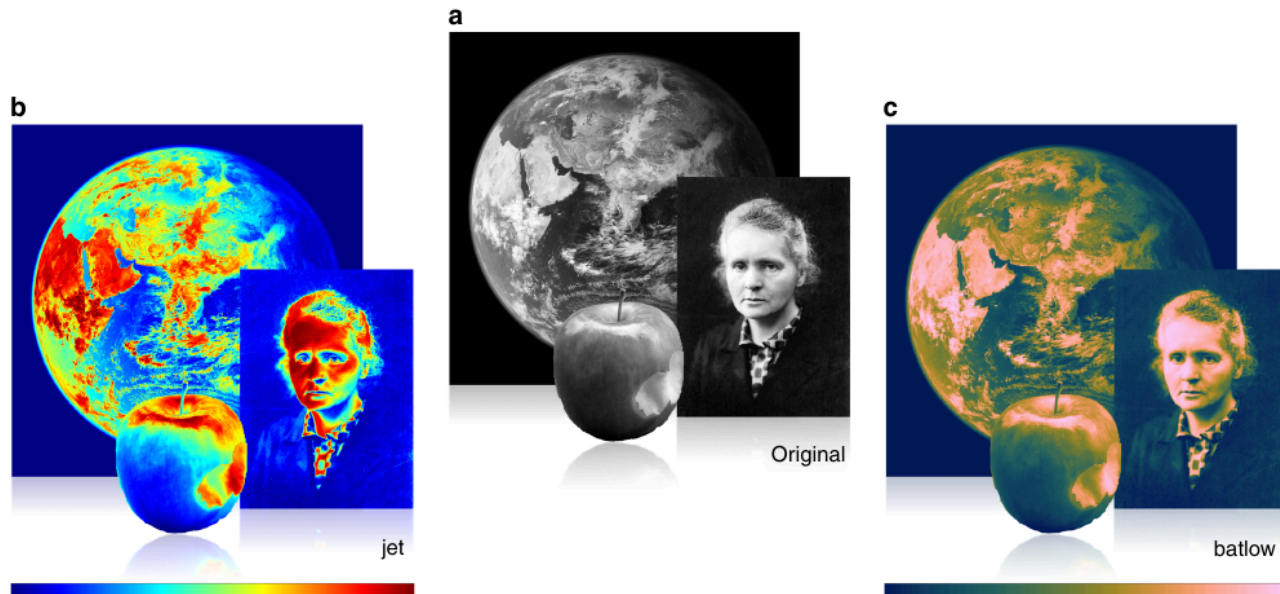




Color vision deficiency affects approximately **1 in 12 men** and **1 in 200 women** worldwide. **Never let colour alone carry what the reader must not miss:** keep the categories few, pick a colourblind-safe palette, and repeat the distinction with position, shape or a label.



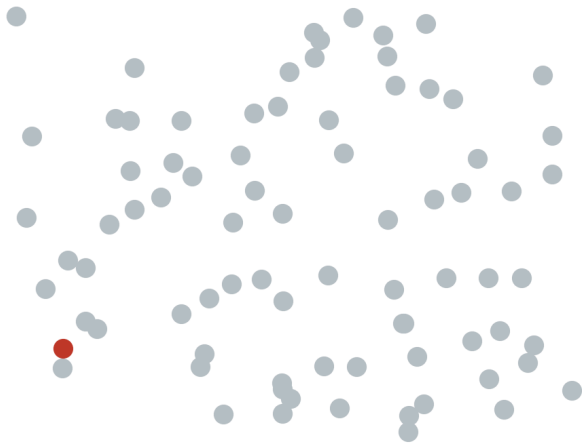
Perceptually uniform colormaps ensure that equal steps in data are perceived as equal steps in color.



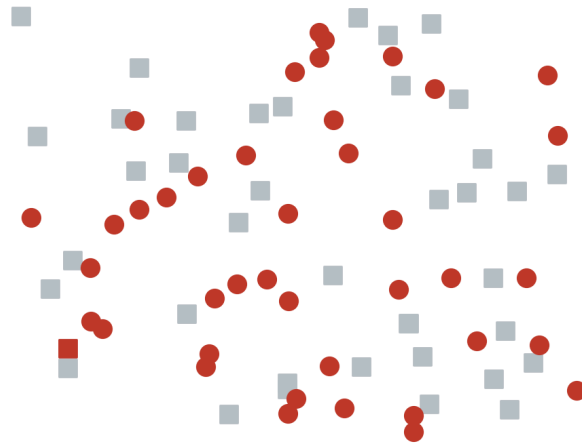
Rainbow colormaps invent structure. The same images as they are (a), in jet (b), and in a perceptually uniform map (c).

Finally, colors can also be used to **draw attention** to specific elements in a visualization, such as highlighting important data points or trends.

One channel: the red point pops out

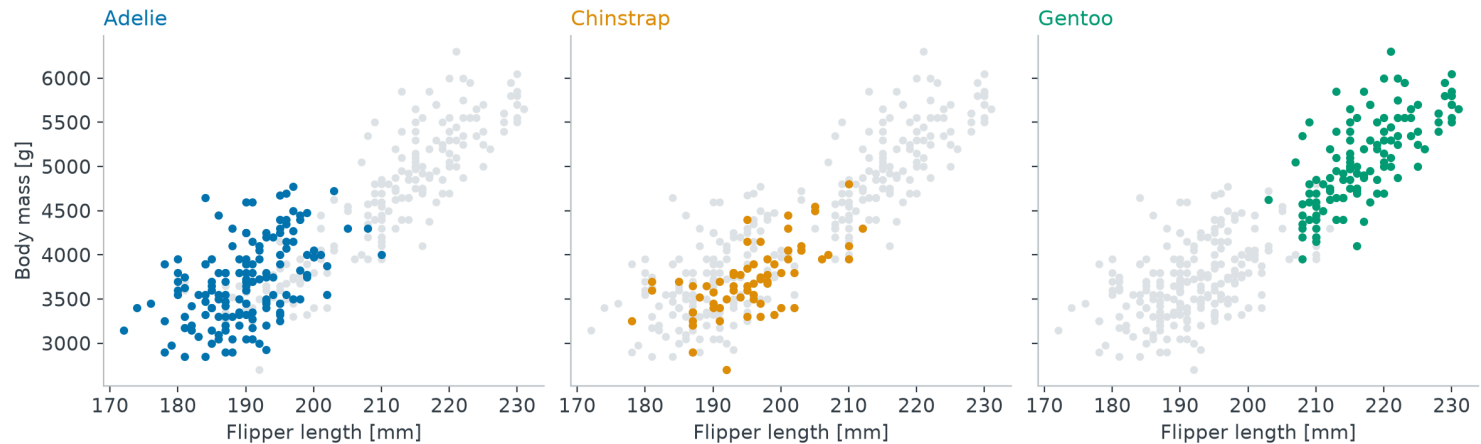


Two channels at once: the red square must be searched



A single channel pops out at a glance; two channels at once must be searched.

Small multiples



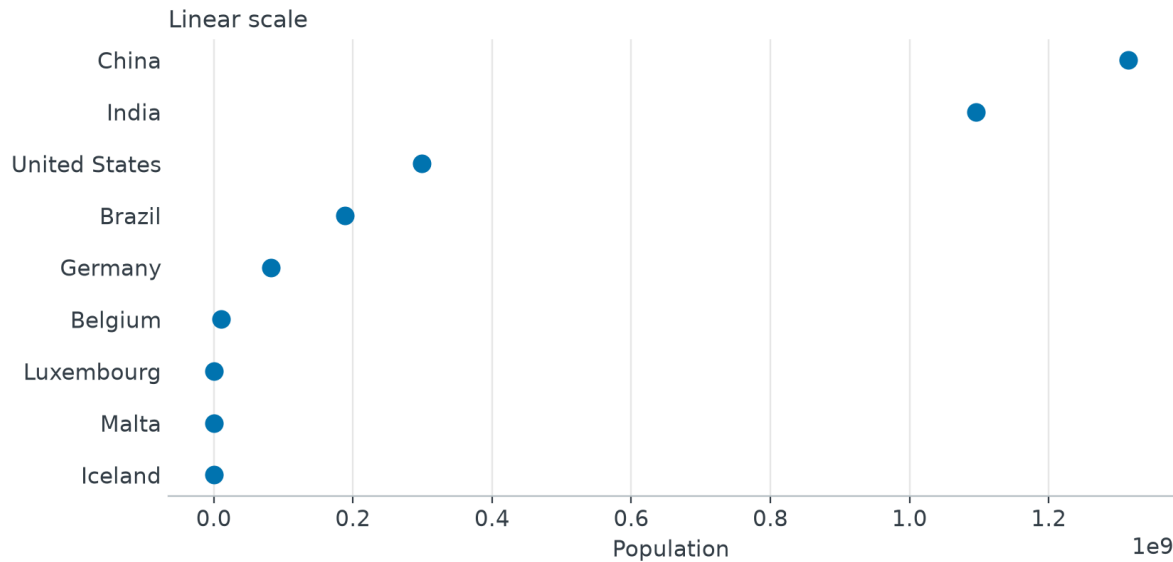
Subsets can be told apart within one panel, by colour, or split into panels sharing their scales. Keeping the other points in grey gives each panel its reference.



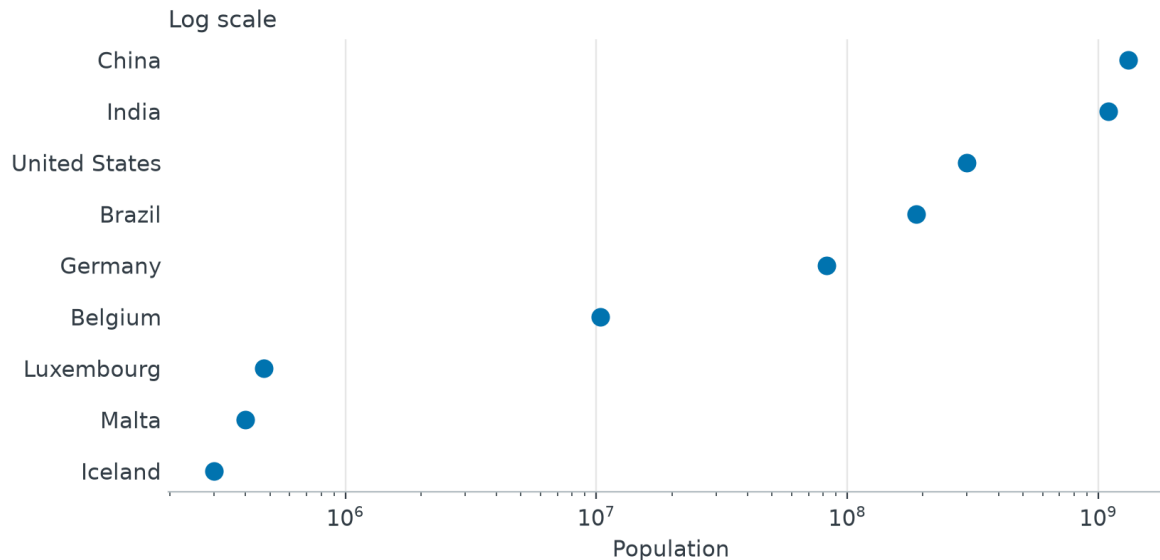
Scales and transformations

Sometimes, data spans several orders of magnitude or has a skewed distribution. In such cases, applying a transformation or using a different scale can improve the interpretability of the visualization.

- **Linear scale:** preserves the original data values.
- **Logarithmic scale:** useful for data spanning several orders of magnitude.
- **Quantile scale:** divides data into bins holding equal numbers of records, useful for skewed distributions.



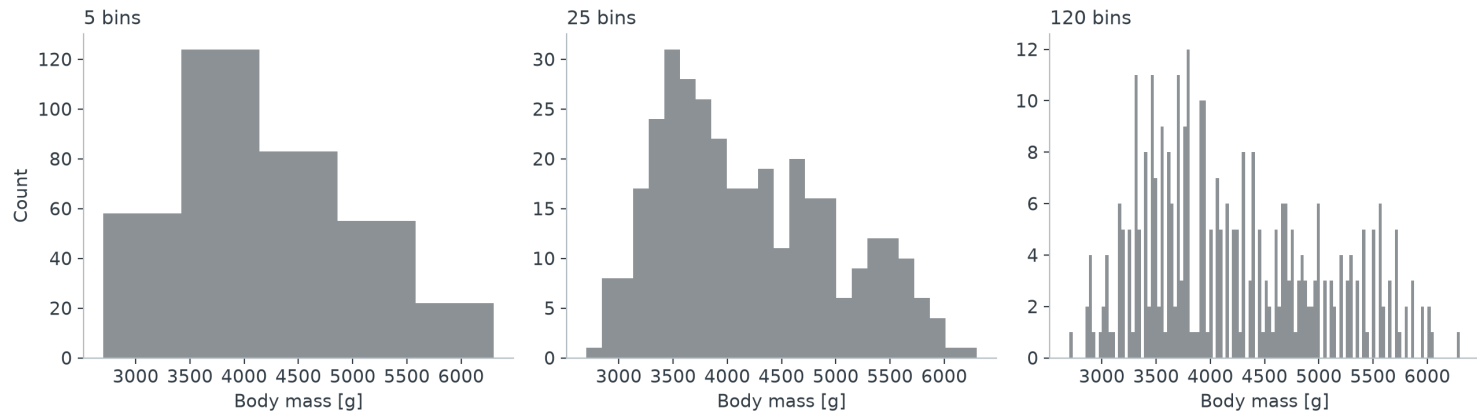
Linear scales can be dominated by large values, dwarfing smaller values and making it hard to see small variations.



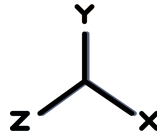
Showing data on a logarithmic scale can prevent large values from dominating the visualization and reveal patterns among smaller values.

It also hides absolute differences, cannot show zero or negative values, and is read as linear by an audience that does not expect it.

Binning is a choice



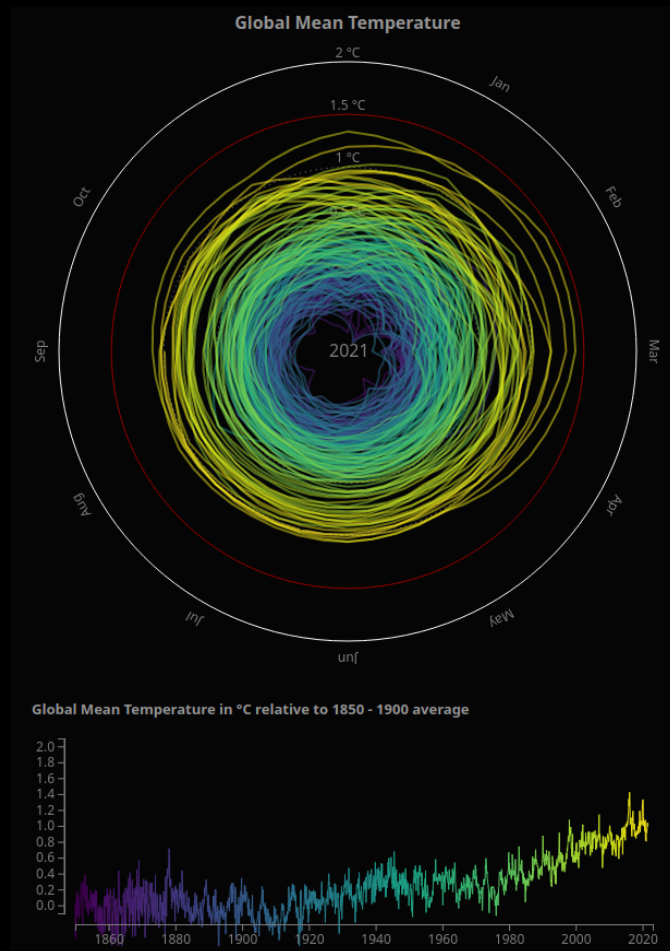
The same 342 body masses. Too few bins hide the second group, too many show noise as structure. Bin width, like kernel bandwidth, belongs to the plot and not to the data.



Coordinate systems

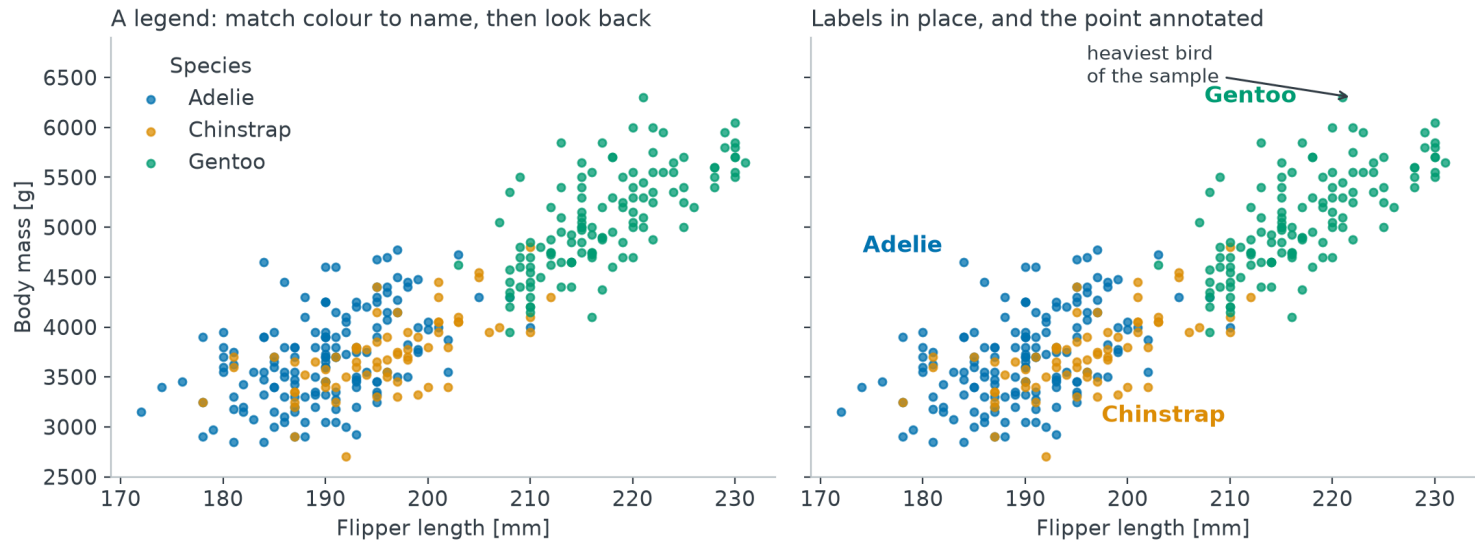
Before mapping (numerical) data to visual channels, it is important to choose an appropriate coordinate system as it affects the perception and effectiveness of the visualization.

- **Cartesian coordinates:** intuitive and effective for most data types.
- **Polar coordinates:** useful for cyclic data (e.g., time of day, seasons), but can distort perception of lengths.



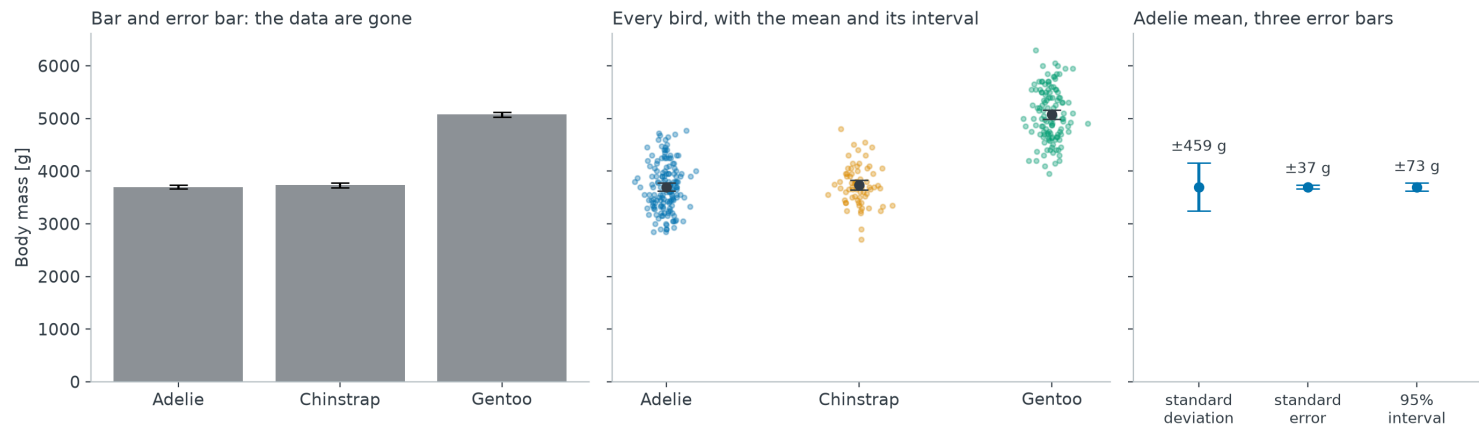
Climate spiral vs. line chart showing global mean temperature change over time.

Guides



A legend costs a lookup for every group. Labelling them in place, and annotating the one thing the reader should notice, costs nothing.

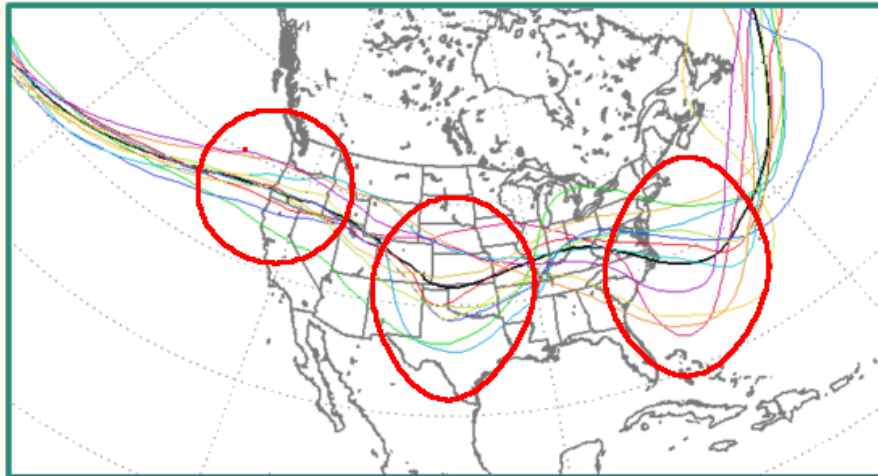
Show the uncertainty



A bar with an error bar shows a mean and a width, and hides everything else. Show the data, then the estimate and its interval.

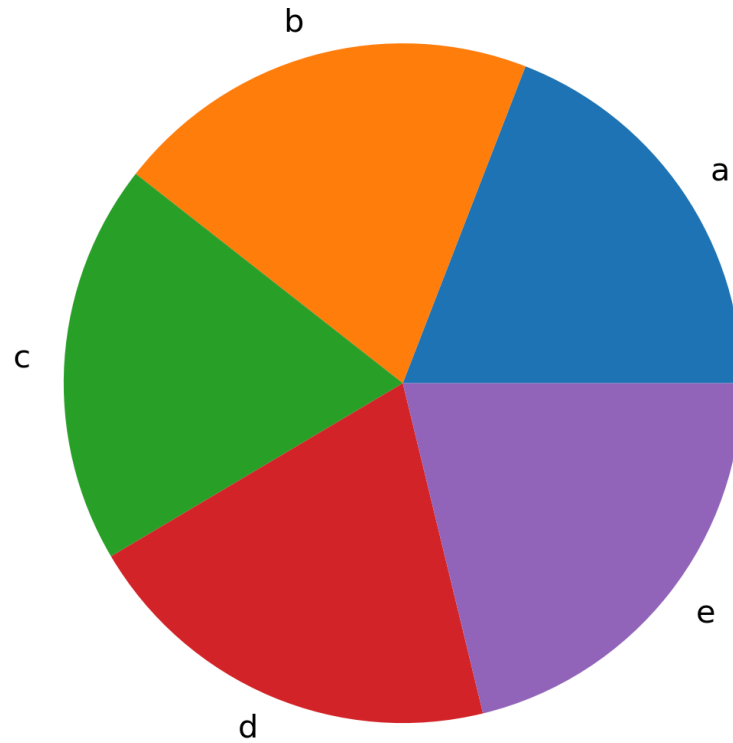
An error bar means nothing until the caption says what it is: here the same Adelie mean carries ± 459 g, ± 37 g or ± 73 g, depending on whether the bar is a standard deviation, a standard error or a 95% interval.

Ensemble mean/members 564-dm 500-hPa height line,
00z 19Nov01 forecast valid 12z22Nov01

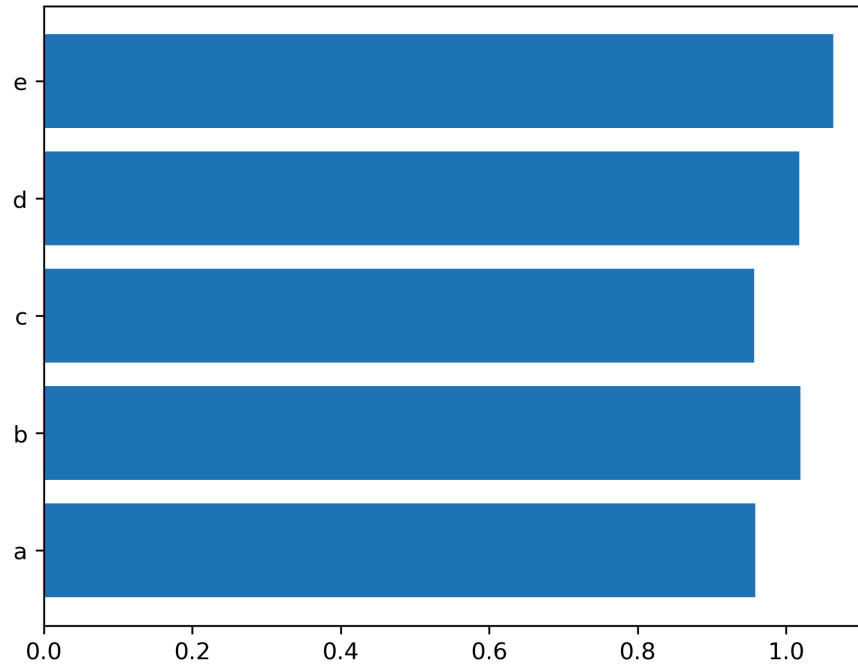


Ten forecasts of the same contour, three days ahead.
Too many lines on purpose: the message is the spread.

Anti-patterns

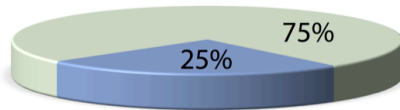


What category is the largest?

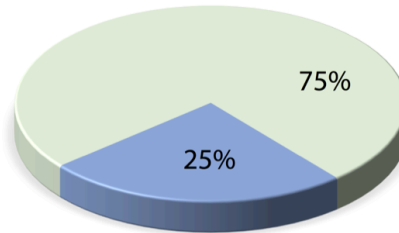


Same data! **Lengths and positions are easier to compare** than angles and areas.

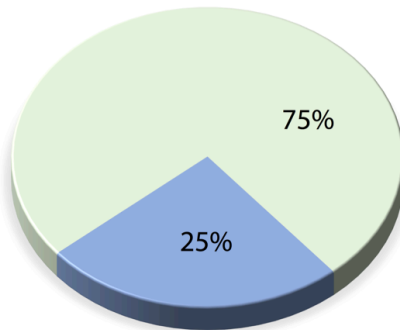
a



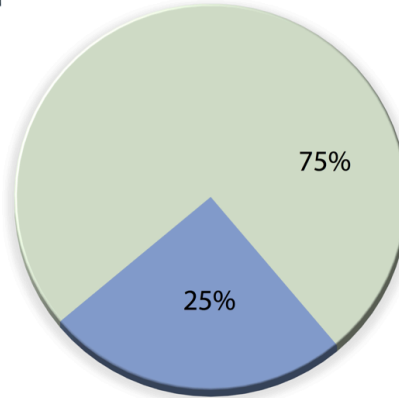
b



c



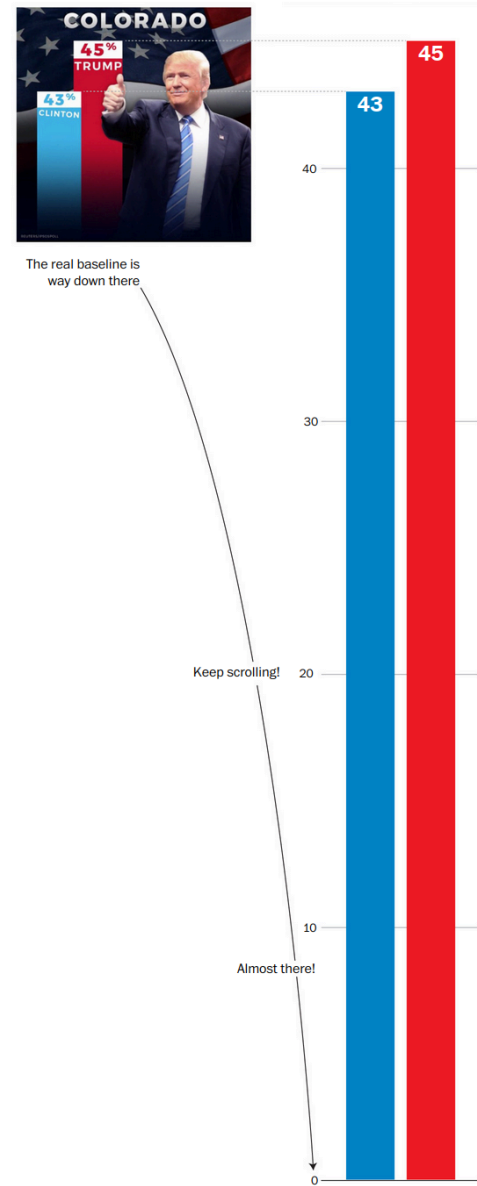
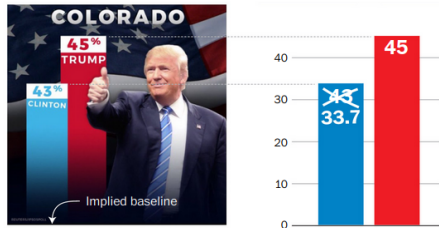
d

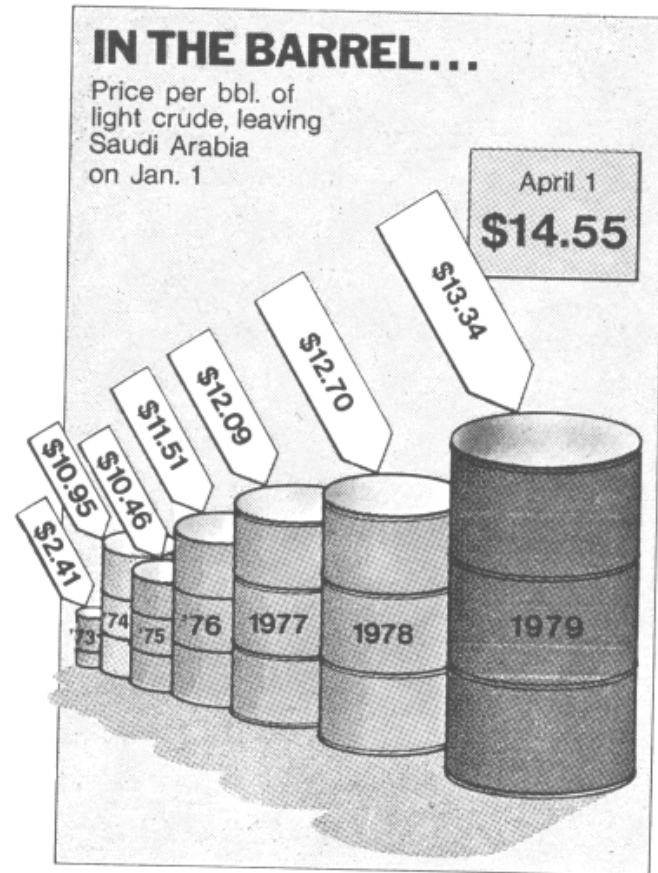
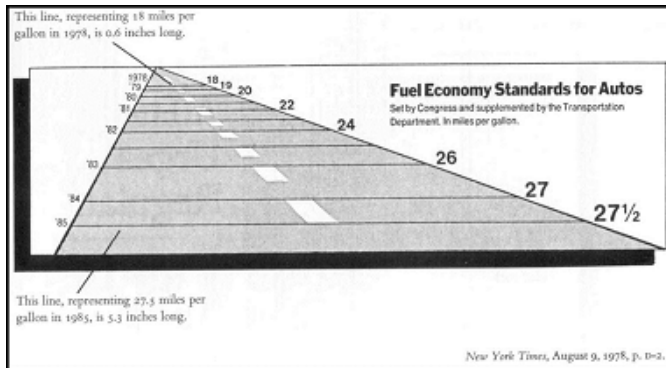


Do not go for 3D. It distorts perception and adds unnecessary complexity.



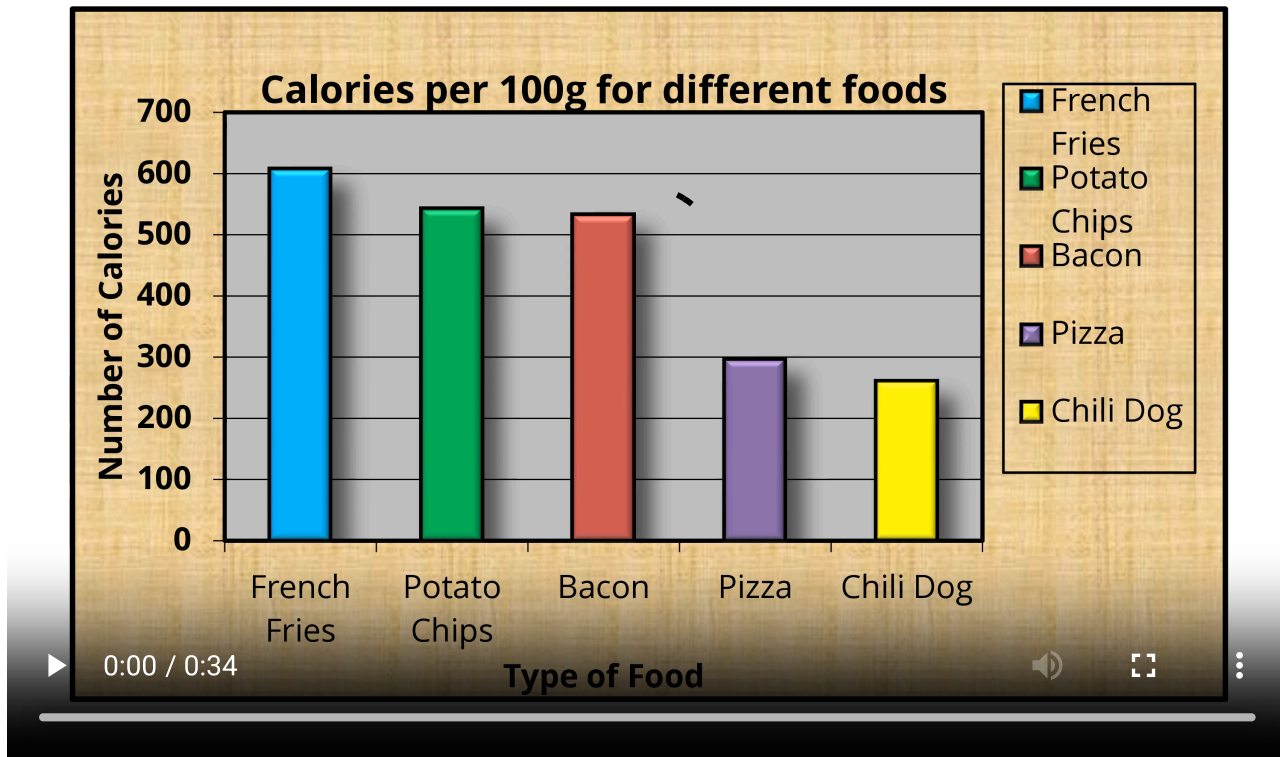
Do not exaggerate reality by truncating axes or using misleading aspect ratios.





Do not lie. Use visual channels that accurately represent the data.

Remove backgrounds



Maximize the data-ink ratio by removing unnecessary elements.

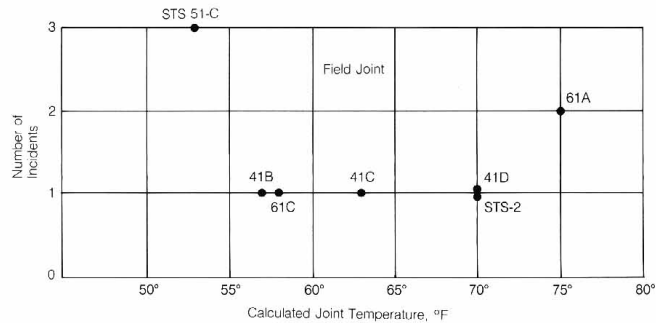


Figure 6
Plot of flights with incidents of O-ring thermal distress as function of temperature

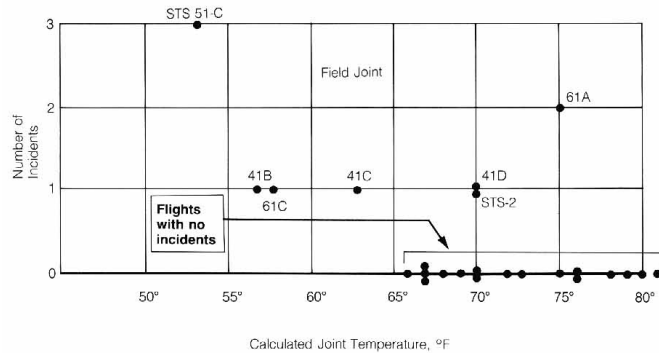


Figure 7
Plot of flights with and without incidents of O-ring thermal distress

NOTE: Thermal distress defined as O-ring erosion, blow-by, or excessive heating

Do not plot a subset. Above, only the flights that had damage; below, all of them.

Wrap-up exercise



Let us discuss the following examples. For each of them, identify what is good and what could be improved.

(All plots are taken from MSc theses of previous years. Author names have been removed to protect the innocent.)

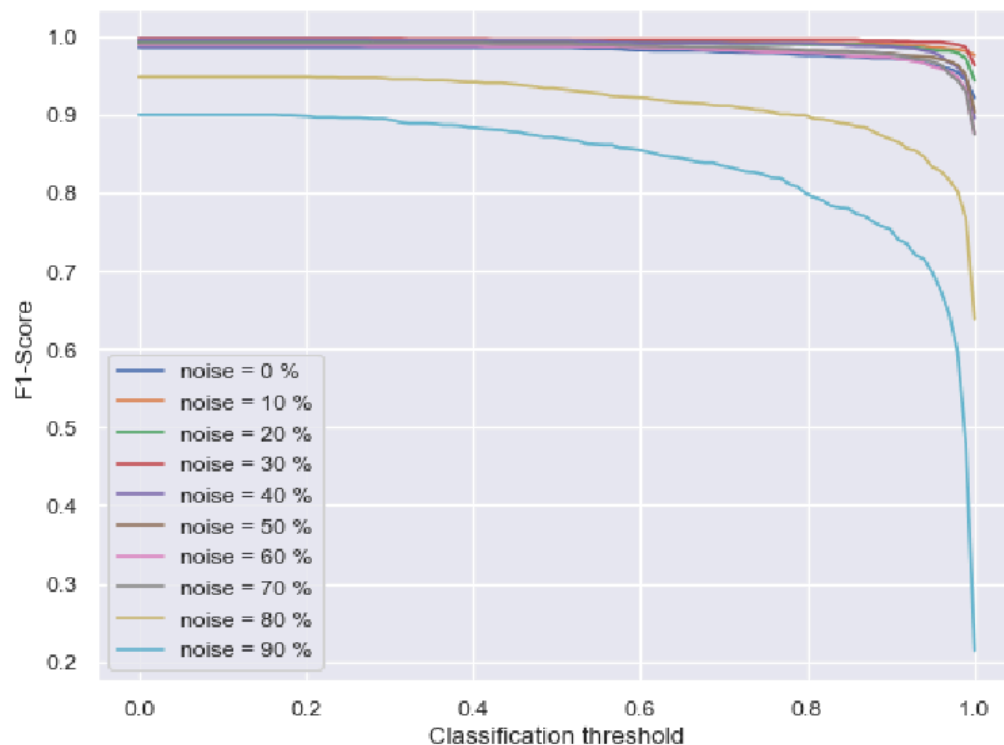


FIGURE 4.21: F1-Score of the best model trained with different quantities of noise going from 0 to 90% and tested on the real conditions dataset according to a classification threshold.

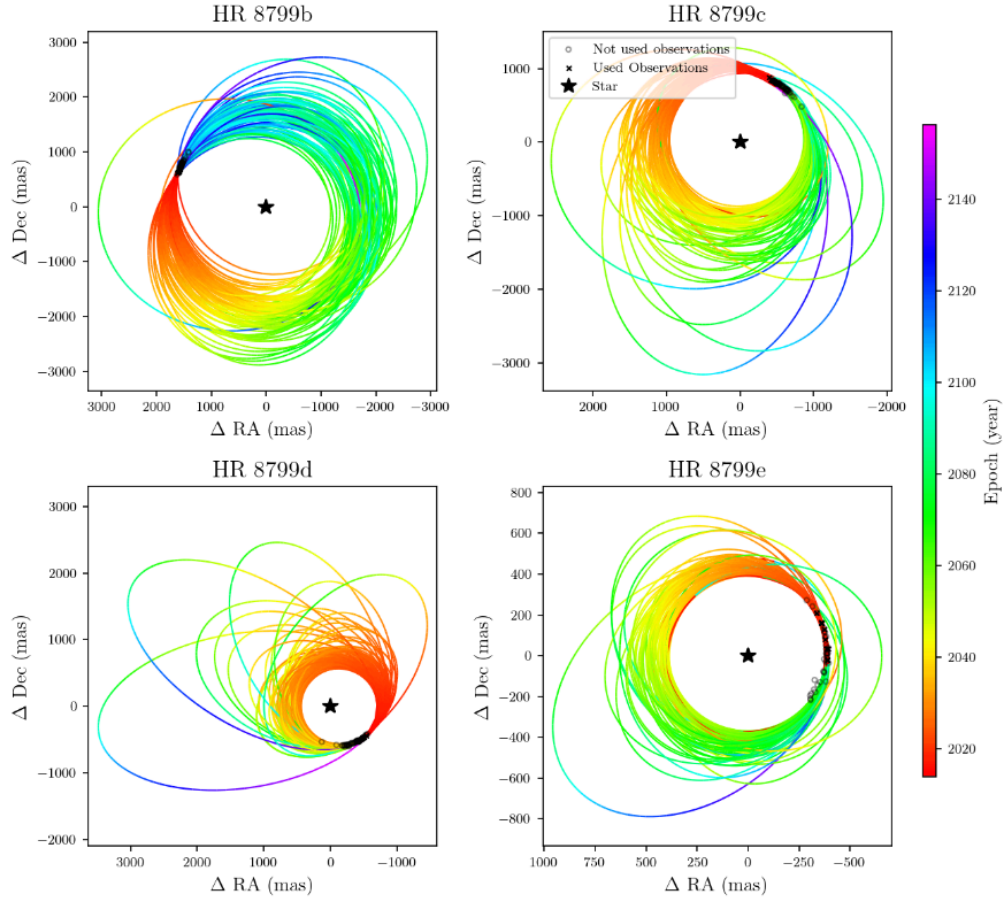
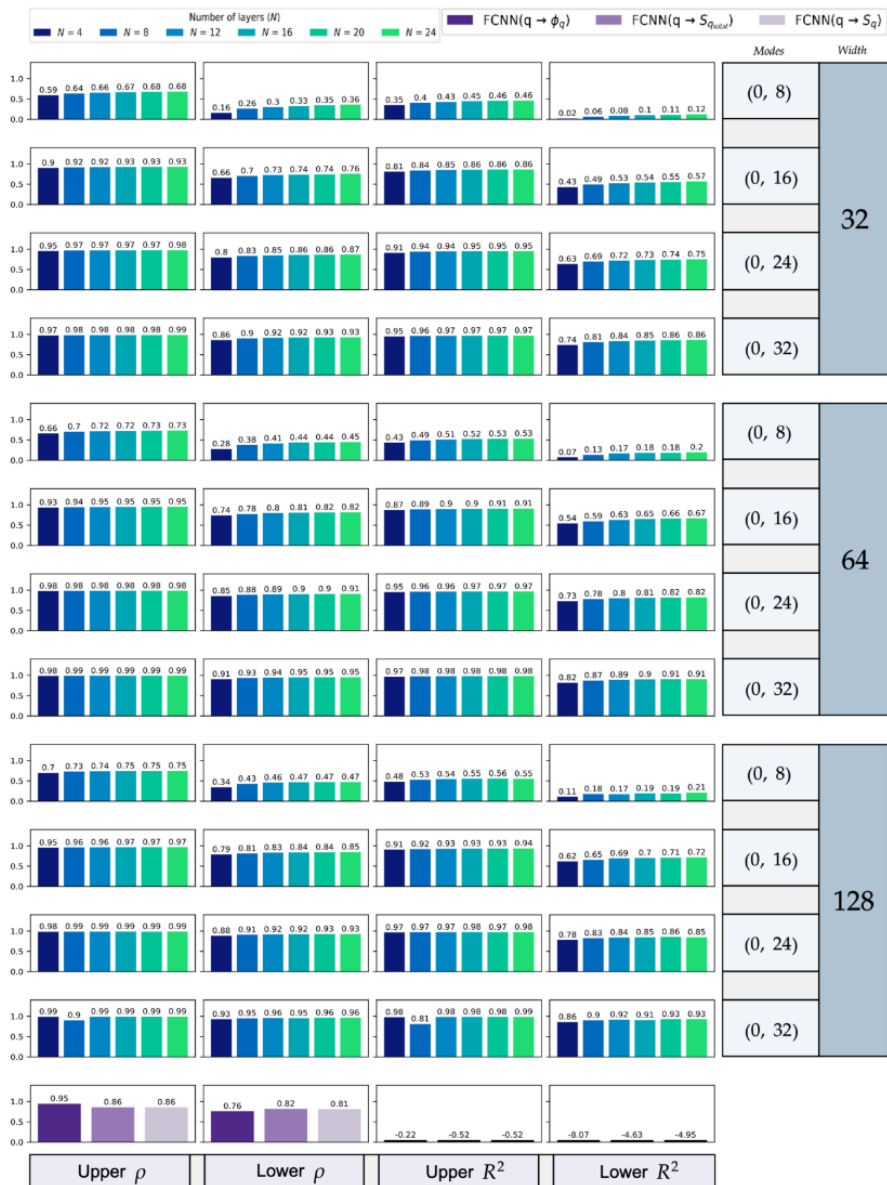


Figure 5.12. Posterior predictive check of the 4 exoplanets of HR 8799, 1000 orbits are generated for each exoplanet. The NPE produces a good approximation of those orbits, however, some impossible orbits are still generated like on HR8799b, HR8799c or HR8799e where there are orbits that are not passing through the observations. The x-axis has been inverted to make the plot comparable with the one in the paper of Sepulveda and Bowler (2022) [40].



Key takeaways

- Start from the message: comparison, distribution, correlation or evolution.
- Encode it with the channels read most accurately: position and length first.
- Pick scales, coordinates and colours that do not distort what the data say.
- Avoid the classics: 3D, bars cut off from zero, colour as the only cue.

A good plot answers a question, and does not let the reader misread the answer.

